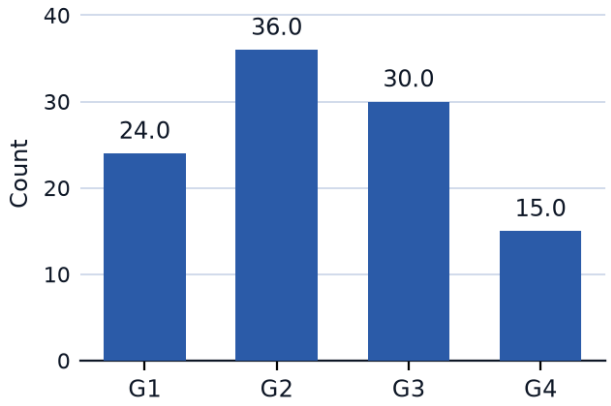
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

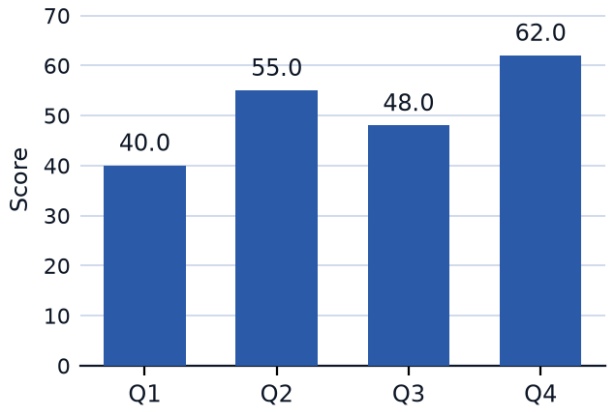
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 4$ in box plots and measures of spread, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 5$ in data distributions, evaluate $f(2)$.



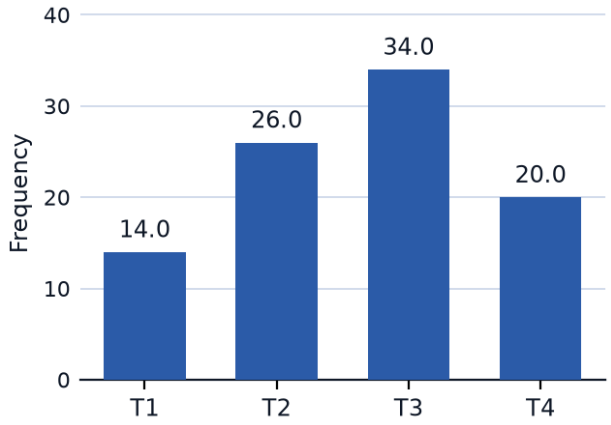
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for data distributions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 6$ in comparisons of data sets, evaluate $f(2)$.

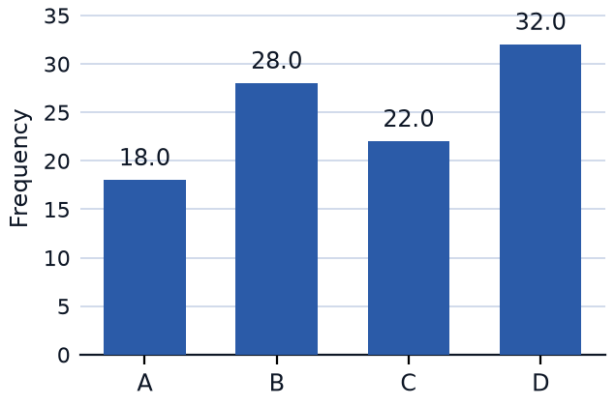


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for comparisons of data sets consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 7$ in two-way frequency tables, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for two-way frequency tables consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In an arithmetic sequence, the 4th term is 17 and the 10th term is 47. Find the first term a_1, the common difference d, and the sum of the first 25 terms S_{25}.

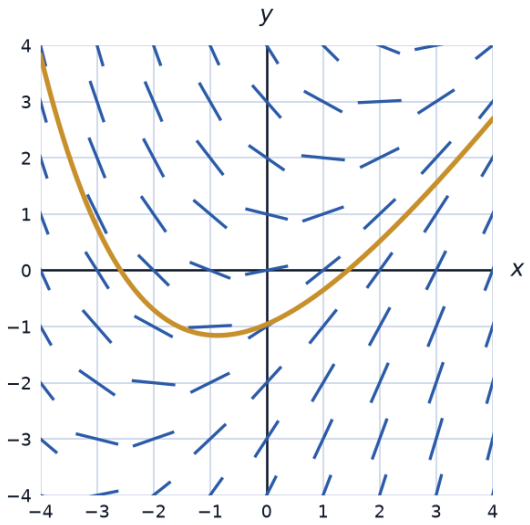
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9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the coefficient of x^2 in the expansion of $(3 - 2x)^5$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the ordered dataset of 12 numbers: 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 38, 72. (a) Determine Q_1, the median, Q_3, and the IQR. (b) Use Tukey's $1.5 \times$ IQR rule to determine whether 72 is an outlier.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An equilateral triangle has a side length of 12 cm. Find: (a) the exact length of its altitude, and (b) its exact area.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Solve $2x^2 - 6x + 5 = 0$ by completing the square, expressing all solutions in exact $a \pm bi$ form.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two assembly line components operate independently. Component A has a 95% reliability rate (probability of functioning properly is 0.95), and Component B has a 90% reliability rate (probability of functioning is 0.90). What is the probability that at least one of the two components fails?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For what value of the constant a will the system of equations $ax + 4y = 7$ and $3x + 2y = 5$ have no solution?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

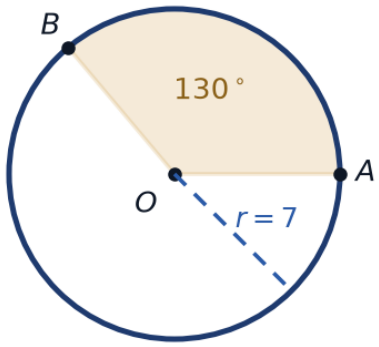
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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Statistics & Data Analysis

Find the equation of the perpendicular bisector of the line segment joining $P(2, -1)$ and $Q(8, 7)$. Give your answer in the form $y = mx + c$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Quadratic Functions

Convert $\frac{7\pi}{6}$ radians to degrees, and find the exact coordinates of its terminal point on the unit circle.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Exponents & Polynomials

A rectangular water storage reservoir has dimensions 2.5 meters by 3 meters by 1.2 meters. A drainage pump empties the reservoir at a constant rate of 75 liters per minute. Given that $1\ \text{m}^3 = 1000$ liters, how many hours will it take to completely empty the full reservoir?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Functions & Transformations

In the system $ax + 4y = 18$ and $3x + 2y = 9$, the system has no solution. What is the value of a?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A line segment connects points $P(-3, 7)$ and $Q(5, -1)$. Determine: (a) the exact length of PQ, (b) the coordinates of the midpoint M, and (c) the slope of the perpendicular bisector of PQ.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Systems & Inequalities

An investment fund gains 30% in value during Year 1, decreases by 20% in Year 2, and increases by 10% in Year 3. Determine the overall net percent change of the fund across the three-year period.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 9$ in frequency data displays, evaluate $f(2)$.



a. 13

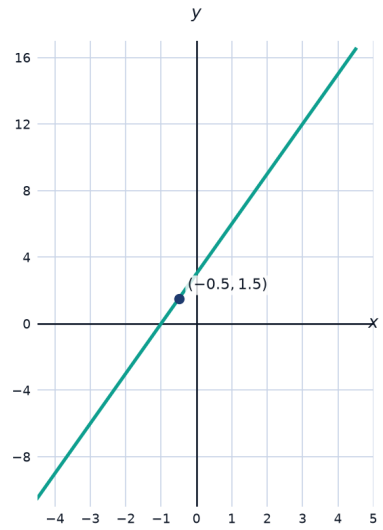
b. 15

c. 4

d. 17

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 3$ in measures of center, evaluate $f(2)$.



a. 9

b. 12

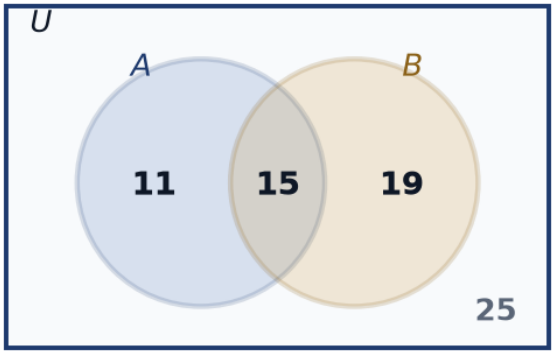
c. 6

d. 13

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for frequency data displays consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?



a. 15.5

b. 16.5

c. 17.5

d. 19.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for measures of center consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Prove by induction that $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$. Step: assume true for n, verify for $n+1$.

a. Not provable by induction

b. Only for $n \le 100$

c. $\frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$. ✓

d. Only for even n

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Prove: in any geometric sequence with positive terms, the geometric mean of two terms equals the intermediate terms.

a. Not always true

b. Only for $r = 1$

c. Use AM-GM

d. If a, ar, ar^2: $\sqrt{a \cdot ar^2} = ar =$ middle term

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference d. The terms a_1, a_3, and a_4 form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the n-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?

a. $\frac{5}{4}$

b. $\frac{81}{64}$

c. $\frac{9}{8}$

d. $\frac{5}{8}$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider an infinite geometric series with first term a and common ratio r, where $|r| < 1$: $a + ar + ar^2 + ar^3 + \ldots = 12$ If each term in the series is squared, the sum of the resulting infinite geometric series is: $a^2 + a^2r^2 + a^2r^4 + a^2r^6 + \ldots = 72$ What is the value of the common ratio, r?

a. $\frac{1}{3}$

b. $\frac{1}{2}$

c. $\frac{2}{3}$

d. $\frac{1}{4}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A geometric sequence has its first term a_1 and common ratio r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. Find the value of a_1 given that $req1$.

a. 1

b. 3

c. 7

d. 9

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

a. 48

b. 72

c. 96

d. 192

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the sequence a_n defined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$ for $n \geq 1$. If the sequence converges to a limit L, find the value of $L^2 - 2L$.

a. 1

b. 2

c. 3

d. 4

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

An arithmetic sequence begins with the terms: $8, 13, 18, 23, \ldots$ What is the 15th term of the sequence?

a. 78

b. 83

c. 73

d. 117

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

An arithmetic sequence has a first term of $a_1 = 5$ and a common difference of $d = 3$. What is the value of the 6th term, a_6?

a. 20

b. 23

c. 17

d. 18

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the arithmetic sequence where the first term $a_1 = 7$ and the common difference $d = -3$. Which of the following is the explicit formula for the nth term of this sequence?

a. $a_n = 7 + 3(n - 1)$

b. $a_n = 7 - 3(n - 1)$

c. $a_n = 3 - 7(n - 1)$

d. $a_n = -3 + 7(n - 1)$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A sequence is defined by the recurrence relation $a_n = 2a_{n-1} - a_{n-2} + 3$ for $n \geq 3$. If the first term $a_1 = 5$ and the second term $a_2 = 8$, find the value of a_4.

a. 17

b. 20

c. 23

d. 26

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $|r| < 1$, prove $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ using the limit of partial sums.

a. Use calculus only

b. Cannot be proved

c. $S_N = \frac{1 - r^{N+1}}{1 - r} \to \frac{1}{1-r}$ as $r^{N+1} \to 0$

d. Only for $0 < r < 1$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Prove $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$ using the formula for arithmetic sums.

a. Cannot be proved simply

b. By FTA

c. By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation

d. Use telescoping

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Functions & Transformations

Find a_1 and d for the arithmetic sequence where $S_{10} = 155$ and $a_{10} = 28$.

a. $a_1 = 3,\ d = \frac{25}{9}$

b. $a_1 = 5,\ d = 3$

c. $a_1 = 1,\ d = 3$

d. $a_1 = 3,\ d = 3$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Prove by induction: $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

a. Not provable

b. Only for even n

c. Use calculus

d. Base: $n = 1$: $1 = 6/6$. ✓. Step: add $(n+1)^2$ and factor.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Systems & Inequalities

Find the closed form for F_n (Fibonacci) using Binet's formula.

a. $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$ where $= 1 + 52,\ = 1 - 52$

b. $F_n = \phi^{n-1}$

c. $F_n = 2^n/5$

d. $F_n = n\phi$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Factoring Polynomials

Using partial fractions, find $\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)}$.

a. 1

b. $\frac{1}{2}$

c. $\frac{n}{2n-1}$

d. $\frac{n}{2n+1}$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Radicals & Exponents

The formula for the sum of n terms of an arithmetic series is:

a. $S_n = \frac{a_1}{1-r}$

b. $S_n = \frac{n}{2}(a_1 + a_n)$

c. $S_n = na_1$

d. $S_n = a_1 \cdot \frac{r^n - 1}{r - 1}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Linear Equations & Lines

A geometric series with $|r| < 1$ has an infinite sum of:

a. $S = a_1 \cdot r$

b. $S = \frac{a_1}{1-r}$

c. Undefined

d. $S = \frac{a_1}{r}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Rational Expressions

Find the first term and common difference of the AP whose pth term is $3p + 2$.

a. $a_1 = 2, d = 3$

b. $a_1 = 5, d = 3$

c. $a_1 = 5, d = 2$

d. $a_1 = 3, d = 2$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Statistics & Data Analysis

Use sigma notation for $1^2 + 2^2 + 3^2 + \cdots + n^2$.

a. $\prod_{k=1}^{n} k^2$

b. $\sum_{k=0}^{n} k^2$

c. $\sum_{k=1}^{n} k^2$

d. n^2

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